

utils

$$\text{Die}_x = \frac{\text{Grid.Width}}{\text{range}(x)}, \quad \text{Die}_y = \frac{\text{Grid.Height}}{\text{range}(y)}$$

$$\begin{aligned} \text{range}(x) &= \max(x) - \min(x) \\ \text{range}(y) &= \max(y) - \min(y) \end{aligned}$$

$$\begin{aligned} x_i' &= x_i - \min(x) \\ y_i' &= \max(y) - y_i \\ \forall(x_i', y_i') &\geq 0 \end{aligned}$$